

THE UNIVERSITY OF
SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering

Tutorial Quiz – 1

Individual Quiz (5%)

Quiz Writing Time: **30 minutes**
Friday, 17 April, at 08 PM

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. A data engineering team is building a pipeline to ingest data from multiple external APIs whose structures frequently change (new fields, missing fields, nested variations). Which design choice is most appropriate?

- ☐ Use a strictly normalised relational schema before ingestion
- ☐ **Use a schema-on-read approach with a document-oriented store**
- ☐ Use a schema-on-write approach with enforced constraints
- ☐ Use a columnar data warehouse with a fixed schema

B. Which statement best describes the role of staging in an OLTP → OLAP pipeline?

- ☐ Staging tables are the final tables used by BI dashboards; fact/dim tables are just temporary
- ☐ Staging replaces the OLTP system by serving user transactions with high concurrency
- ☐ Staging exists only to store aggregated results (e.g., monthly revenue)
- ☐ **Staging stores a raw or lightly cleaned copy of source data to support reruns, deduplication, and incremental loading before building fact/dim tables**

C. You can either (i) run SQL that filters/aggregates in the DB and return a small result set, or (ii) bulk-load the whole table into Pandas and process it there. Which is the strongest technical reason for choosing option (i)?

- ☐ Pandas cannot do joins or group-bys
- ☐ SQL cannot express aggregations compared to Pandas
- ☐ Bulk-loading is always slower, even for small datasets
- ☐ **DBMS can optimise queries, and bulk-loading requires the full dataset to fit into memory**

Part II – Short Answer Questions

Provide answers in the provided boxes.

Scenario 1: A fact table grows to billions of rows. Most queries filter by date and region, but performance and cost are rising.

Question 1: What strategies would you use to improve performance and control cost (e.g., partitioning, aggregates), and how do you decide which to implement? **(3 Marks)**

Define metrics centrally (data catalogue/semantic layer) with agreed filters (e.g., only status='Shipped'). Publish tables and prevent ad-hoc redefinition by creating views. Add data tests (for accepted statuses and revenue formula checks) and documentation.

Scenario 2: Your analysts want near-real-time dashboards. The OLTP database is already under heavy load. Running large analytical queries against it slows down customer operations.

Question 2: What extraction approach would you choose (see options below), and how would you architect the system so analytical workloads do not impact OLTP performance? Choose one choice below and justify your selection. **(2 Marks)**

1. Change Data Capture (CDC), i.e., as soon as the data changes
2. Replication
3. Scheduled extracts

1. I would avoid querying the OLTP database directly for dashboards because analytical scans and joins can compete with transactional workloads (locks, CPU, I/O). Instead, I would use CDC (reading the database's transaction log) to stream changes into an analytics environment.
2. - CDC reads changes from the transaction log rather than running large joins/aggregations on OLTP tables.
3. - OLTP already writes to logs for recovery; CDC just consumes those logs.

Scenario 3: You are given a webpage from an online unit allocation, presented in an HTML structure below. You need to extract this data for a data warehouse.

Answer the following:

1. Identify the attributes and write at least one record **(1 Mark)**
2. Which elements will you target for extraction? **(1 Mark)**

```
<!DOCTYPE html>
<html>
<head>
<title>Unit Allocations</title>
</head>
<body>

<div id="courses">
  <div class="course">
    <h2 class="title">Data Engineering</h2>
    <span class="code">COMP5339</span>
    <span class="lecturer">Imdad</span>
    <span class="enrolments">450</span>
  </div>

  <div class="course">
    <h2 class="title">Machine Learning</h2>
    <span class="code">COMP5310</span>
    <span class="lecturer">Lee</span>
    <span class="enrolments">200</span>
  </div>

  <div class="course">
    <h2 class="title">Databases</h2>
    <span class="code">COMP9123</span>
    <span class="lecturer">Imdad</span>
    <span class="enrolments">870</span>
  </div>
</div>

</body>
</html>
```

Attributes: title, code, lecturer, enrolments

Record: Data Engineering, COMP5339, Imdad, 450

Element to target for extraction: div id = "course"